Orbit Decay AI Engine

Research & Implementation Methods

Research Methods

Aggregation and Physics Modeling

To accurately predict orbital decay, we constructed a robust ML pipeline driven by fundamental astrodynamics:

- **1. Historical Aggregation:** Correlated historical UWE-4 TLE data against CelesTrak `SW-All.csv` solar weather data.
- **2. Physics Modeling:** Integrated NRLMSISE-00 to calculate localized exospheric temperature and density.
- **3. Feature Engineering:** Computed 7-Day & 30-Day F10.7 Solar Flux rolling averages, Kp Max geomagnetic activity indices, and extracted ballistic coefficients (B^*).

We implemented an ensemble model to regress drag parameters directly to altitude drops.

Random Forest Regressor

Goal: Predict 7-day and 30-day semi-major axis degradation.

Logic: Map environmental parameters (F10.7, Kp, density) to observed historical orbital drops.

Result: A highly accurate prediction of orbital decay with confidence intervals.

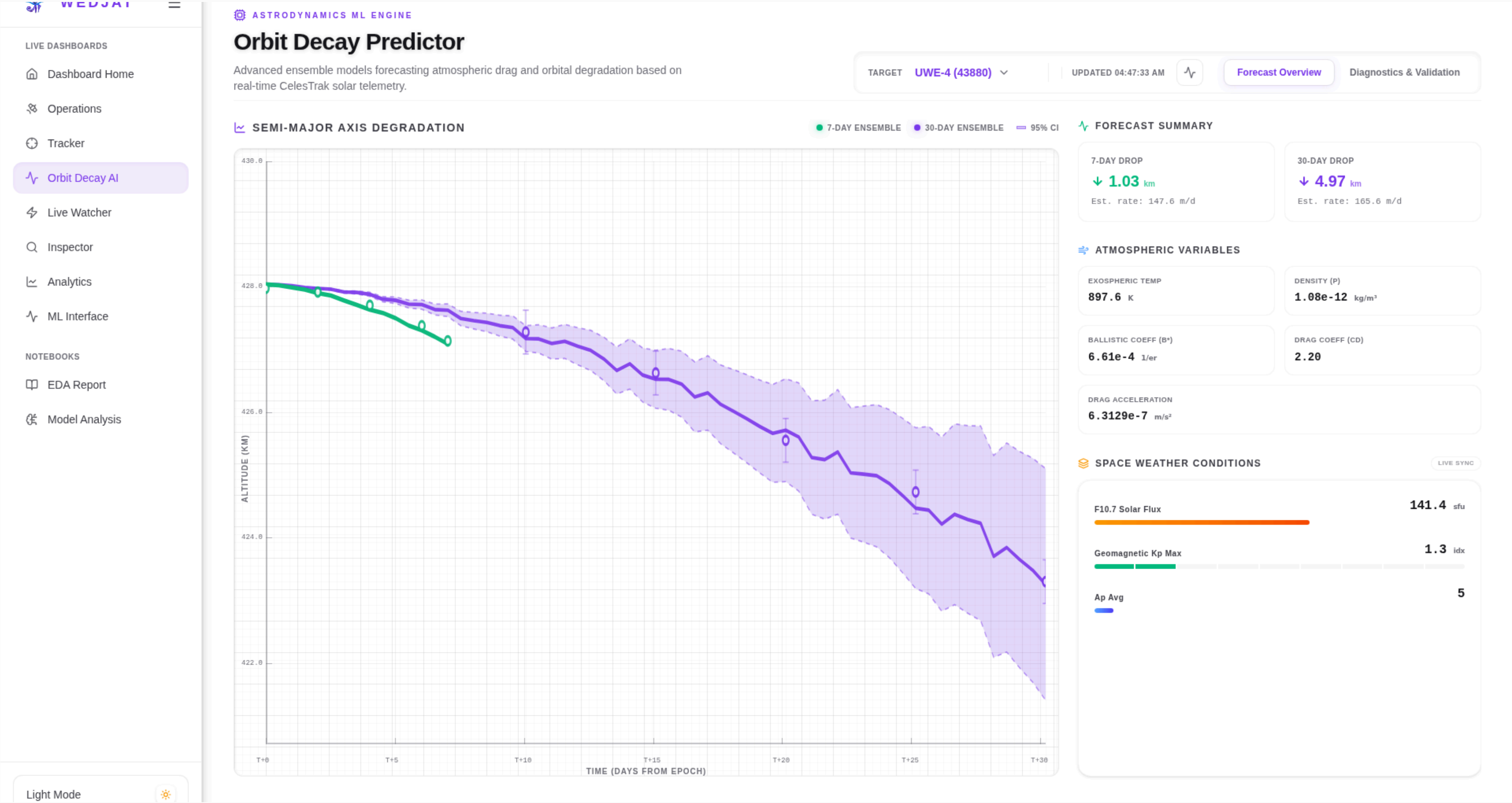
Deployment & Resiliency

The Production Pipeline

The AI models have been integrated into our production FastAPI stack:

- **1. FastAPI Endpoint:** Provides on-demand 7D and 30D forecast inferences.
- **2. Memory Caching:** All CelesTrak outgoing API calls are wrapped in an aggressive 3600s `@ttl_cache` to prevent rate-limiting.
- **3. Environment Locking:** Strict adherence to `scikit-learn==1.6.1` across Pixi and Docker to prevent deserialization faults.
- **4. Svelte UI:** A seamless, reactive dashboard utilizing WebSockets and interactive charts.

Results



LIVE DASHBOARDS
Dashboard Home
Operations
Tracker
Orbit Decay AI
Live Watcher
Inspector
Analytics
ML Interface
NOTEBOOKS
EDA Report
Model Analysis
Light Mode
ASTRODYNAMICS ML ENGINE
Orbit Decay Predictor
Advanced ensemble models forecasting atmospheric drag and orbital degradation based on real-time CelesTrak solar telemetry.
TARGET
UWE-4 (43880)
UPDATED 04:47:33 AM
Forecast Overview
Diagnostics & Validation
SEMI-MAJOR AXIS DEGRADATION
7-DAY ENSEMBLE
30-DAY ENSEMBLE
95% CI
430.0
428.0
426.0
424.0
422.0
ALTITUDE (KM)
T+0
T+5
T+10
T+15
T+20
T+25
T+30
TIME (DAYS FROM EPOCH)
FORECAST SUMMARY
7-DAY DROP
1.03 km
Est. rate: 147.6 m/d
30-DAY DROP
4.97 km
Est. rate: 165.6 m/d
ATMOSPHERIC VARIABLES
EXOSPHERIC TEMP
897.6 K
DENSITY (P)
1.08e-12 kg/m³
BALLISTIC COEFF (B*)
6.61e-4 1/er
DRAG COEFF (CD)
2.20
DRAG ACCELERATION
6.3129e-7 m/s²
SPACE WEATHER CONDITIONS
LIVE SYNC
F10.7 Solar Flux
141.4 sfu
Geomagnetic Kp Max
1.3 idx
Ap Avg
5

Model Diagnostics (Light)